

Title: Design a distributed application using RPC for remote computation where client submits an integer value to the server and server calculates factorial and returns the result to the client program. give me a proper code for this practical in python.

Requirement:

1. Intel i3 , 3.3GHz , 4gb RAM
2. Network Environment
3. Python 3.x.
4. RCP library, xmlrpc.server
5. Math Library

Theory:

This practical exercise demonstrates the principles of distributed computing and Remote Procedure Calls (RPC) using the specific example of factorial calculation. Here's a breakdown of the underlying theory:

a. Distributed Computing

Distributed computing involves dividing a computational task into smaller parts and executing those parts on multiple interconnected computers (nodes).

Client-Server Architecture

This practical employs a client-server architecture, a common model in distributed systems. The server provides a service (factorial calculation), and the client requests that service.

b. Remote Procedure Call (RPC)

RPC allows a program running on one computer to execute a procedure (function) on another computer as if it were a local procedure call.

Mechanism

- The client makes a function call, which is translated into a network request.
- The request is sent to the server.
- The server executes the function.
- The result is sent back to the client.
- The client receives the result and continues execution

XML-RPC

XML-RPC is a simple RPC protocol that uses XML for encoding requests and responses. It relies on HTTP for transport. Its simplicity makes it relatively easy to implement and use.

c. Factorial Calculation

The factorial of a non-negative integer 'n', denoted as $n!$, is the product of all positive integers less than or equal to 'n'.

- $n! = n * (n-1) * (n-2) * \dots * 2 * 1$
- $0! = 1$ (by definition)

Conclusion:

The practical exercise successfully demonstrated the implementation of a distributed application for factorial calculation using XML-RPC. The client program was able to remotely invoke the factorial calculation function residing on the server, sending integer inputs and receiving the corresponding factorial results.

Code:

Server.py

```
Go Run Terminal Help < -> DC prac
server.py x factorial.py client.py
pract1 > server.py > ...
3
4 if __name__ == '__main__':
5     server = xmlrpc.server.SimpleXMLRPCServer(('localhost', 8000))
6     print("Listening on port 8000...")
7     server.register_function(factorial.calculate_factorial, 'factorial') #use imported function
8     server.serve_forever()
```

Factorial.py

```
Go Run Terminal Help < -> DC prac
server.py x factorial.py x client.py
pract1 > factorial.py > calculate_factorial
1 import math
2
3 def calculate_factorial(n):
4     """Calculates the factorial of a given integer."""
5     if not isinstance(n, int):
6         return "Error: Input must be an integer."
7     if n < 0:
8         return "Error: Factorial is not defined for negative numbers."
9     try:
10         return math.factorial(n)
11     except OverflowError:
12         return "Error: Integer too large for factorial calculation"
```

Client.py

```
Go Run Terminal Help < -> DC prac
server.py x factorial.py x client.py x
pract1 > client.py > get_factorial
22 user_input = input("Invalid input. Type 'exit' to quit, or enter an integer: ")
23 if user_input.lower() == 'exit':
24     break
25 else:
26     print("Invalid input")
27
28 except KeyboardInterrupt:
29     print("\nExiting...")
30     break
31 except Exception as e:
32     print(f"An unexpected error occurred: {e}")
33     break
34
```

Result:



The screenshot shows a VS Code terminal window with the 'TERMINAL' tab selected. The terminal displays the command to run a Python script and its output. The file explorer on the right shows the file 'python' selected.

```
PS C:\Users\Rishik Sinha\Desktop\DC prac> & "C:/Users/Rishik Sinha/AppData/Local/Programs/Python/Python313/python.exe" "c:/Users/Rishik Sinha/Desktop/DC prac/prac/c1/server.py"
Listening on port 8000...
127.0.0.1 - - [23/Mar/2025 10:53:44] "POST /RPC2 HTTP/1.1" 200 -

```

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The screenshot shows a VS Code terminal window with the 'TERMINAL' tab selected. The terminal displays the command to run a Python script and its output. The file explorer on the right shows the file 'python' selected.

```
PS C:\Users\Rishik Sinha\Desktop\DC prac> & "C:/Users/Rishik Sinha/AppData/Local/Programs/Python/Python313/python.exe" "c:/Users/Rishik Sinha/Desktop/DC prac/prac/c1/client.py"
Enter an integer to calculate its factorial (or type 'exit' to quit): 10
Factorial of 10: 3628800
Enter an integer to calculate its factorial (or type 'exit' to quit): 

```

Ln 7, Col 45 Spaces: 4 UTF-8 CRLF {} Python 3.13.2